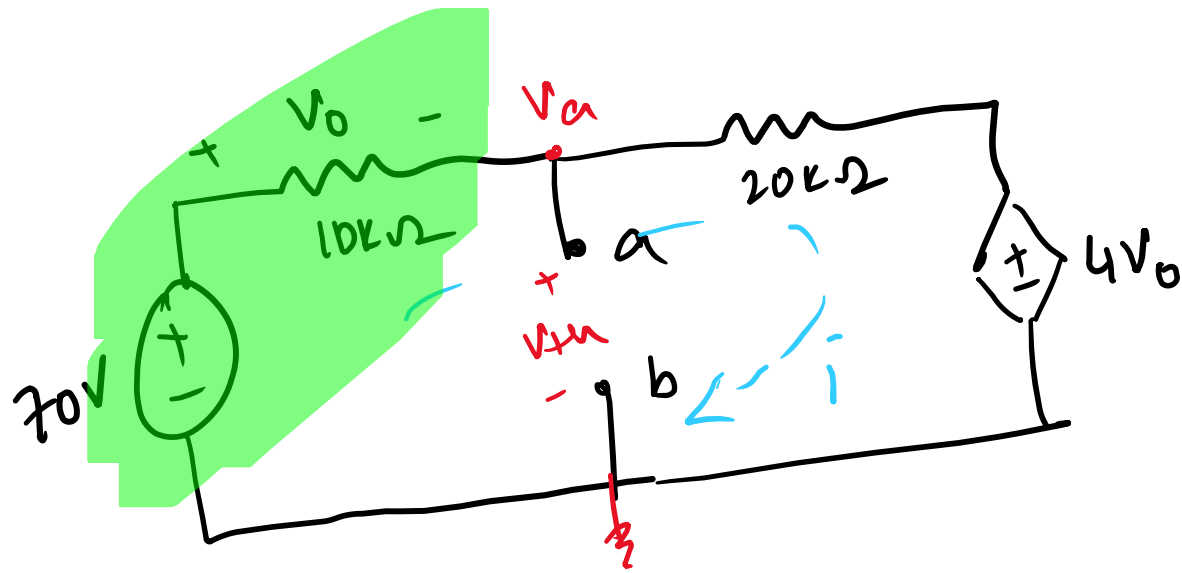




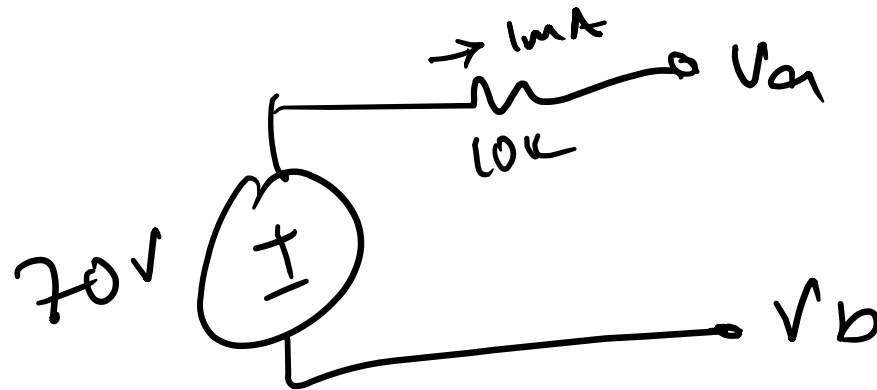
Find Thevenin equivalent at a-b.

$$V_0 = 10ki$$

$$70 - 10ki - 20ki - 4V_0 = 0$$

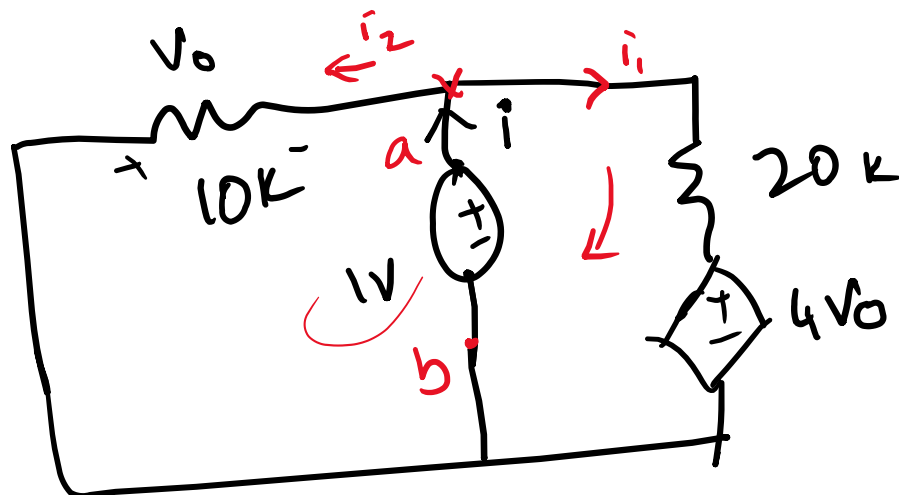
$$70 - 70ki = 0 \Rightarrow i = \underline{\underline{1mA}}$$

$$V_{th} = V_{ab} = 70 - 10ki = 70 - 10 = \underline{\underline{60V}}$$



$$V_{ab} = 70 - 10ki$$

$R_{th}$ !



$$R_{th} = \frac{1V}{i}$$

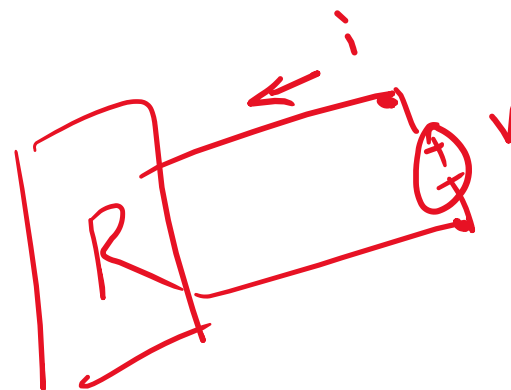
$$V_0 = -1V$$

Mesh :  $1 - 20k i_1 - 4(-1) = 0$   
 $i_1 = 0.25 \text{ mA}$

$$\begin{aligned} \hat{i} &= \hat{i}_1 + \hat{i}_2 \\ &= 0.25 + 0.1 \\ &= 0.35 \text{ mA} \end{aligned}$$

$$R_{th} = \frac{1V}{i} = \underline{\underline{2.857 \text{ k}\Omega}}$$

$$R = \frac{V}{i}$$



Find Thevenin equivalent  
at a-b.

$$V_{th} = 0V$$

(No independent voltage or  
current source)

$$R_{th} = \frac{V_{ab}}{1mA}$$

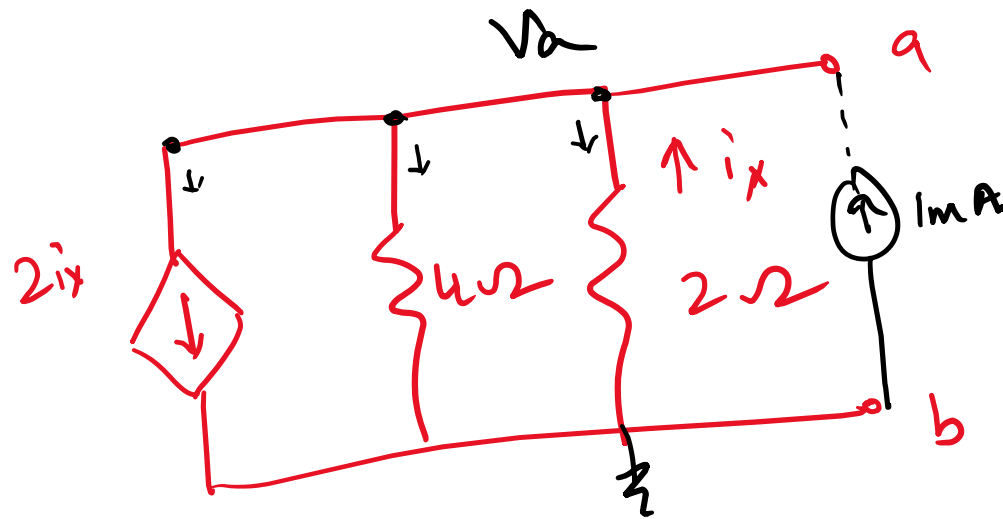
$$\text{At } a: 1mA = \frac{V_a}{2} + \frac{V_a}{4} + 2\hat{i}_x$$

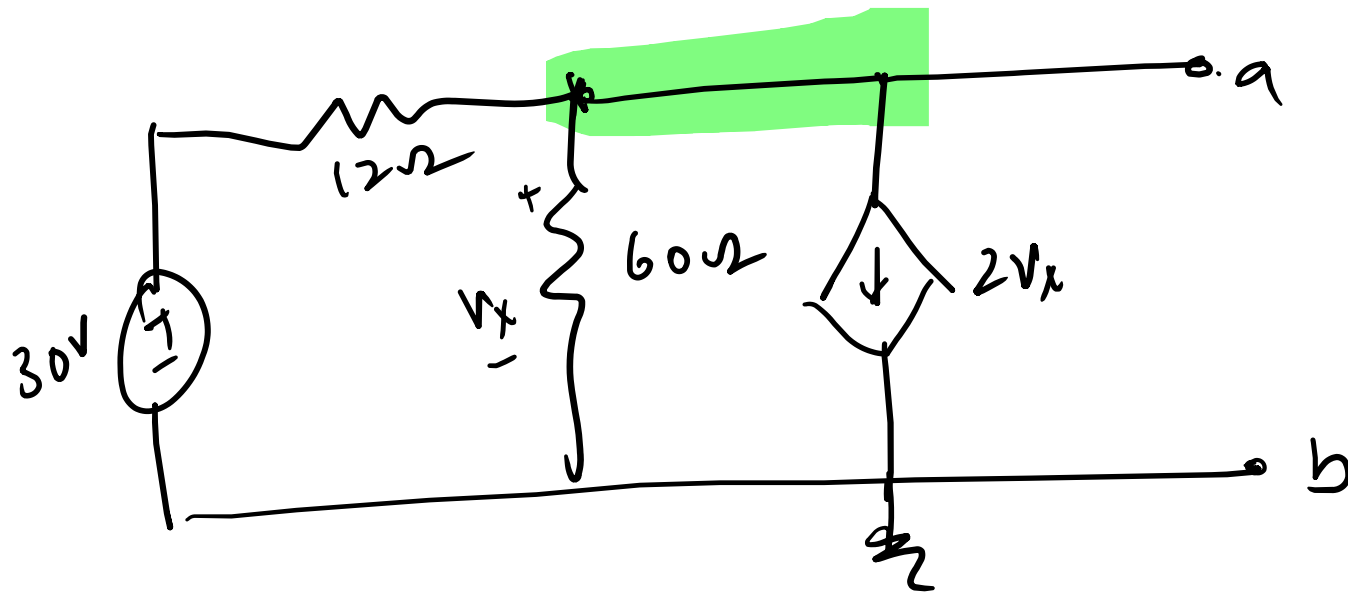
$$\hat{i}_x = -\frac{V_a}{2}$$

$$V_a = -4mV$$

$$R_{th} = \underline{\underline{-4\Omega}}$$

Negative  $R_{th} \rightarrow$  Active device which  
supplies power.



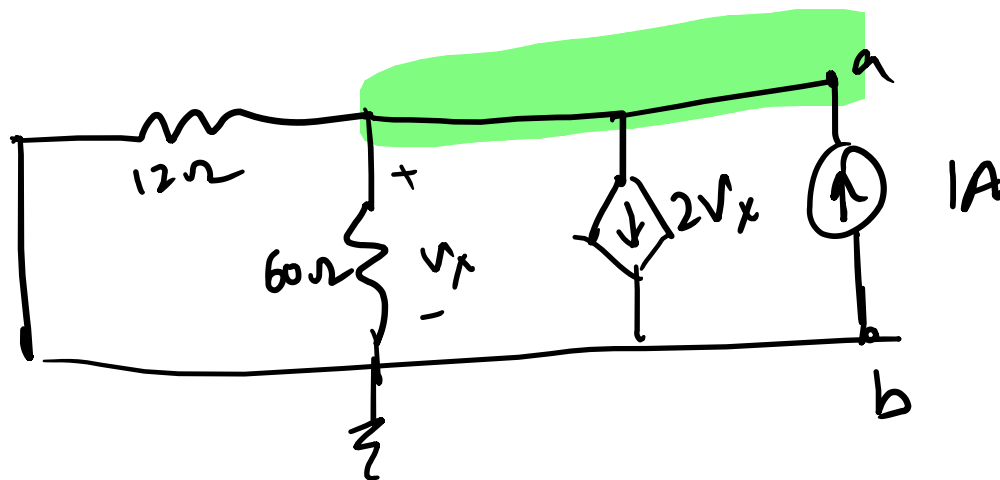


Find Thevenin eq  
at a-b.

$$V_x = V_a = V_{th}$$

$$V_a = V_{th} = 1.19V$$

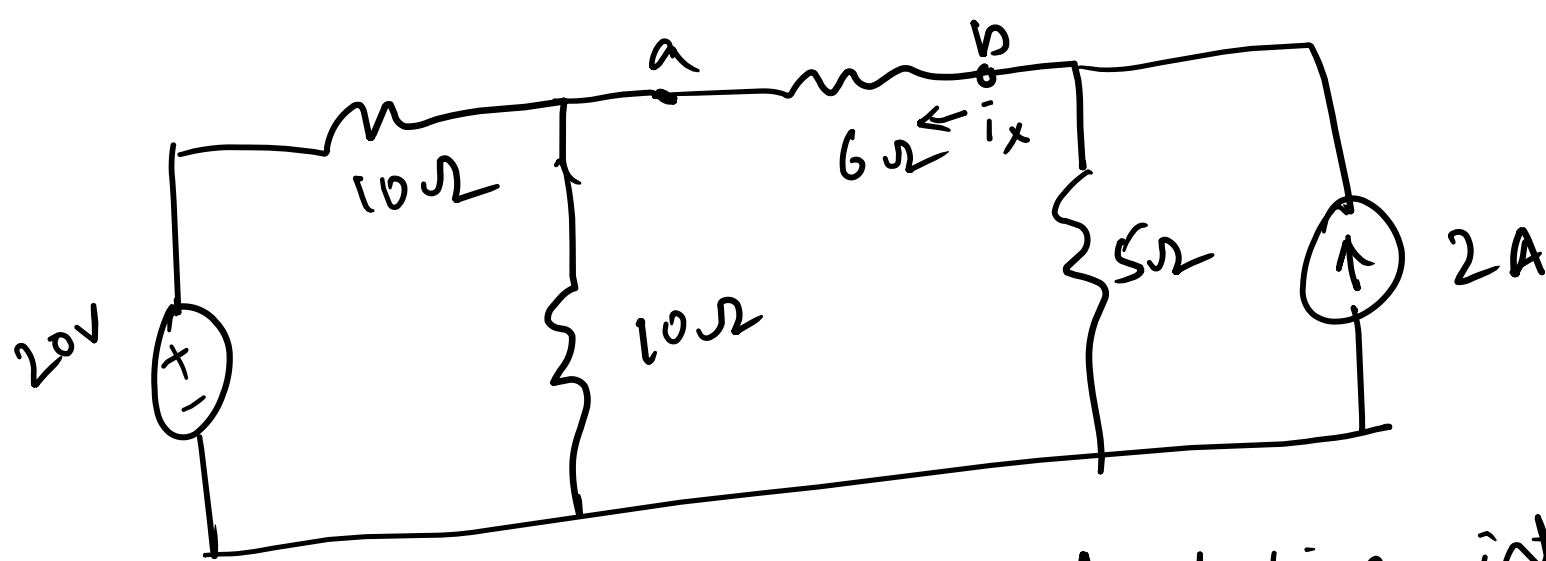
$$\frac{30 - V_a}{12} = \frac{V_a}{60} + 2V_a$$



$$1 = 2V_x + \frac{V_x}{12} + \frac{V_x}{60}$$

$$V_x = 0.476V$$

$$R_{th} = \frac{V_x}{1A} = \underline{\underline{0.476\Omega}}$$



Find the Thevenin equivalent looking into a-b and  
solve for  $i_x$